

CROP SECURITY FROM INTRUDERS

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Problem Statement

- Potent risk for farming is damage to crops by stray/ wild animals that are “INTRUDERS”.
- Loss is not only Economical but also Emotional.
- A System is need to be developed such that Intruders should get repelled from farm without getting harmed.
- All this emphasizes importance of

**“CROP SECURITY FROM
INTRUDERS”**

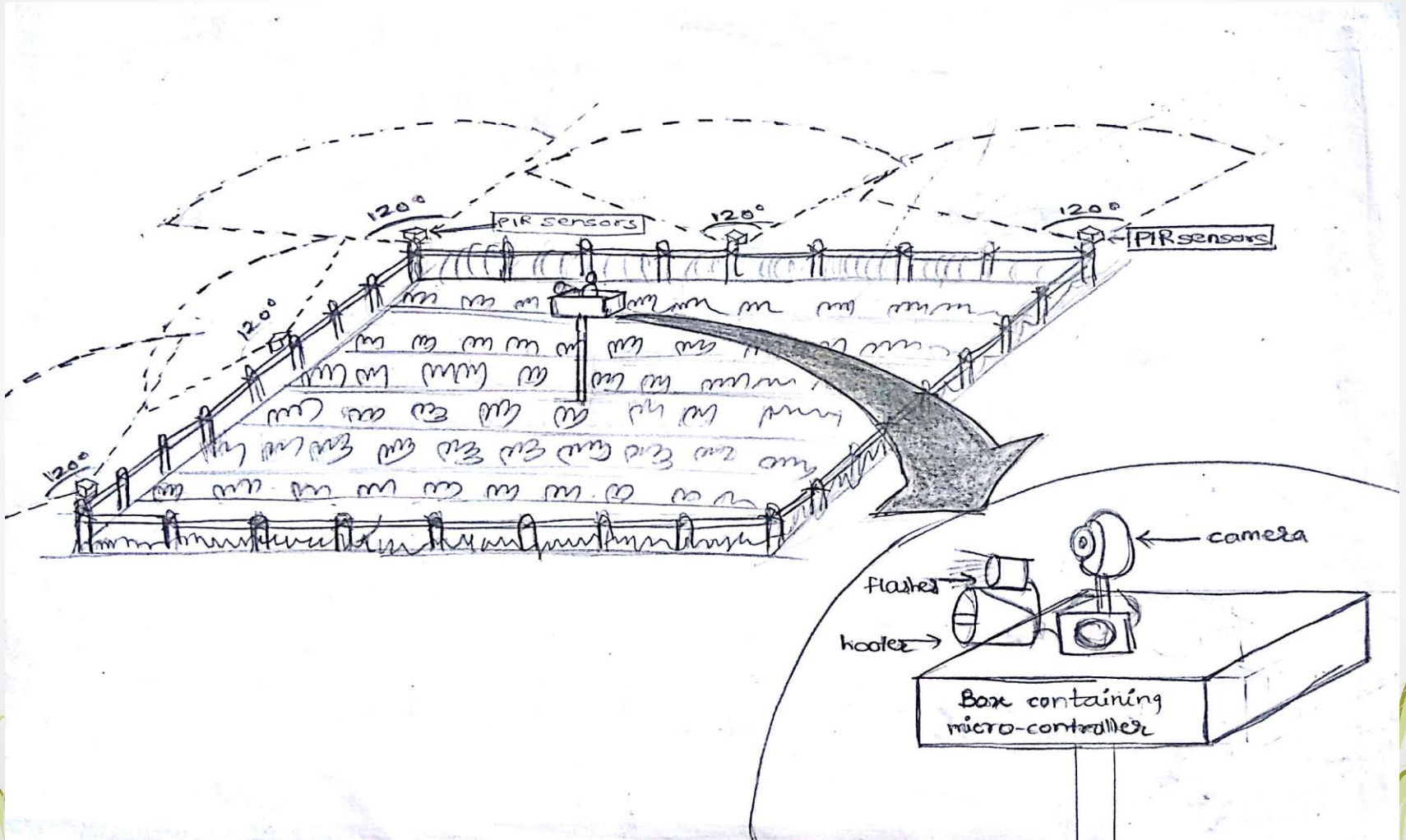


Present practice of crop protection

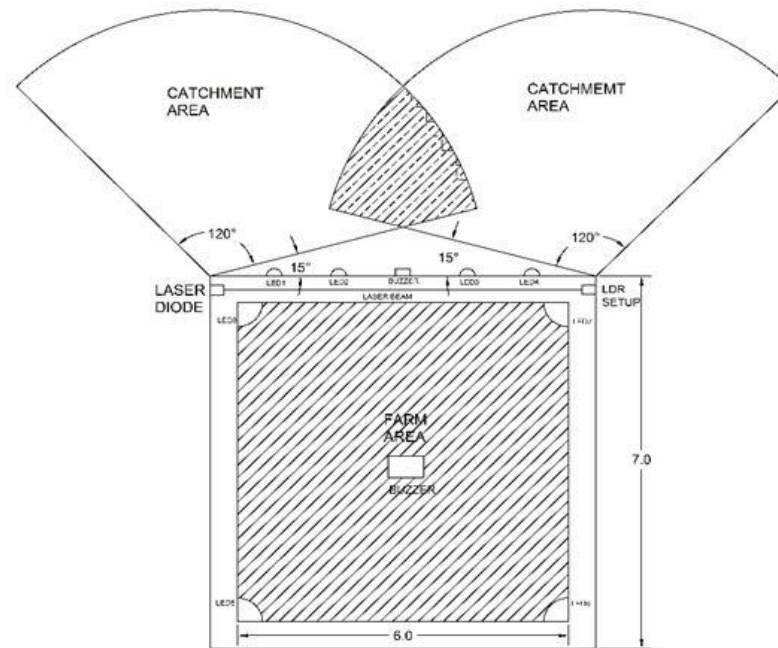
- Manually guarding the fields:
- Employing helpers
- No guarantee of 100% protection of crops
- Electric fence causes harm to intruders



General Approach



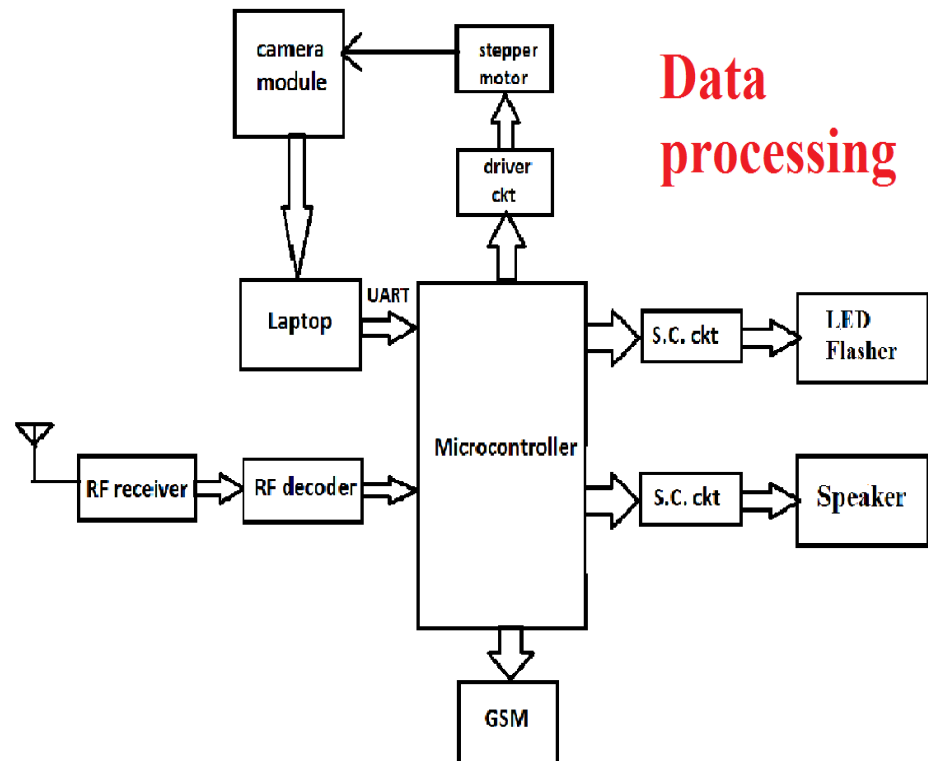
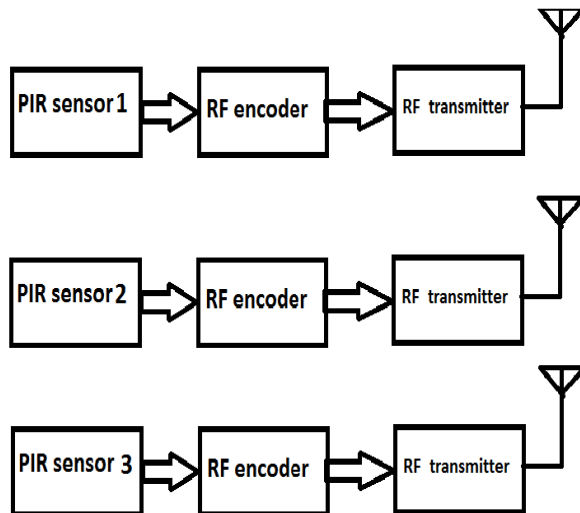
WORKING DIAGRAM



Implementation

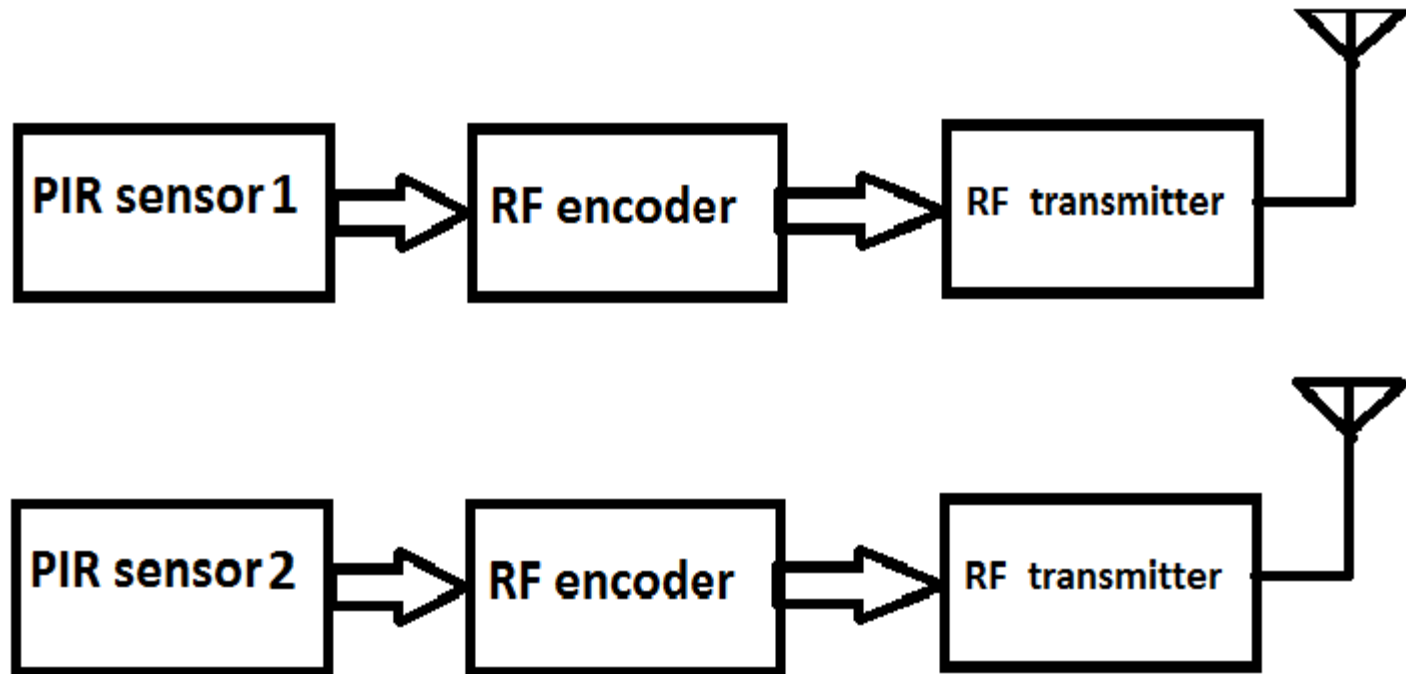
Block diagram

Data Aquisition



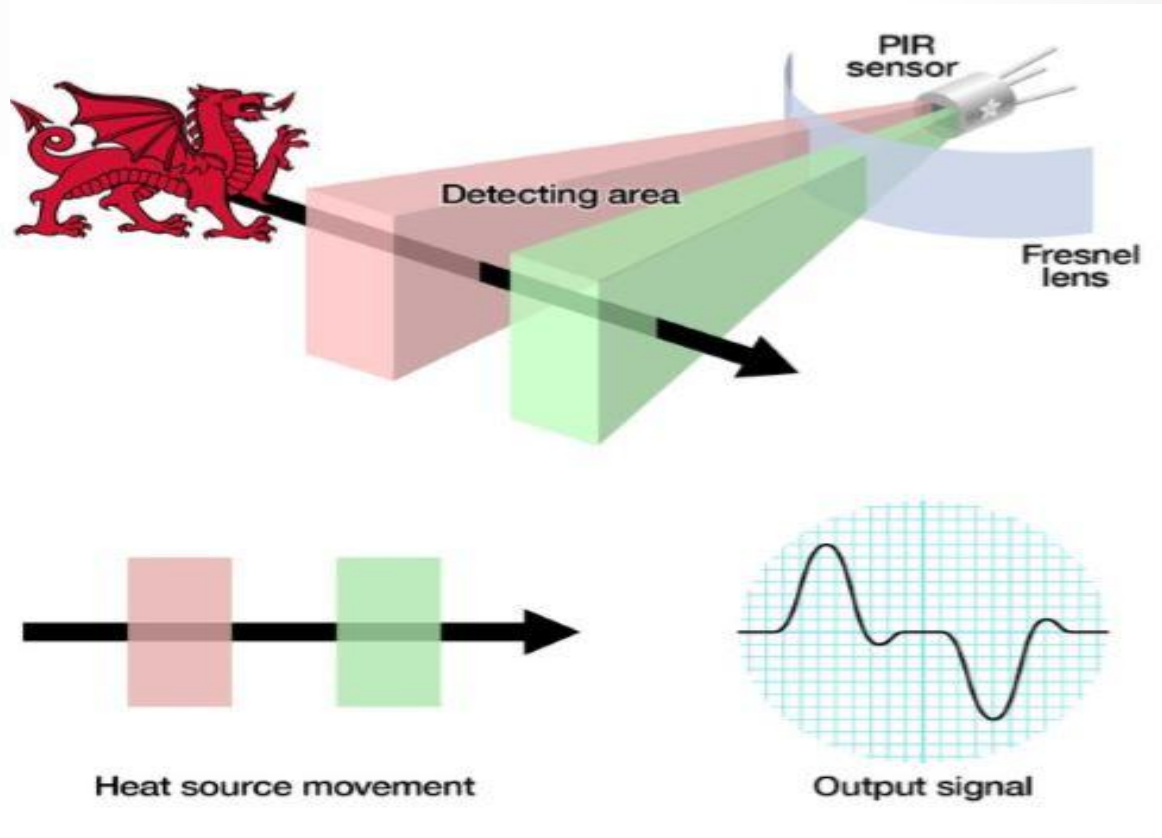
Data processing

DATA ACQUISITION



Two nodes of the PIR sensor module placed at two corners of a farm boundary

PIR Sensor



RF Encoder



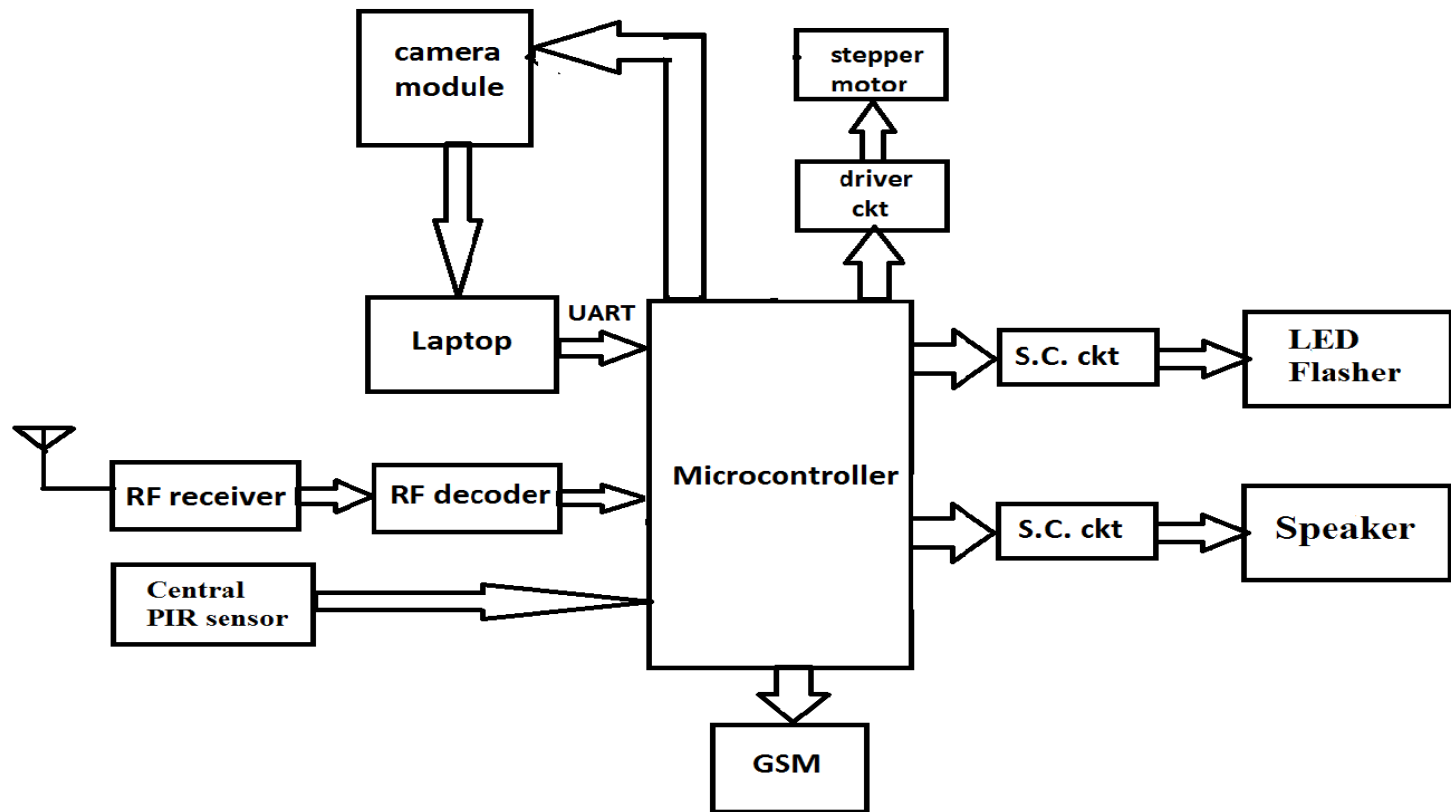
RF Transmitter



TRANSMITTER

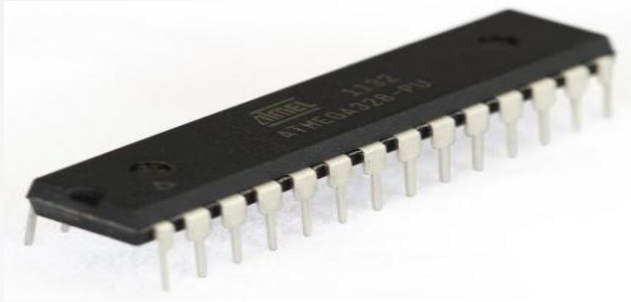
Helps to send the acquired data to the controller

DATA PROCESSING



It situated almost at the centre of the farm to monitor intrusion inside the farm with a camera and PIR module enclosed in a box.

Atmega 328p



GSM



Camera



They form the core part of central unit.

Speaker



LED



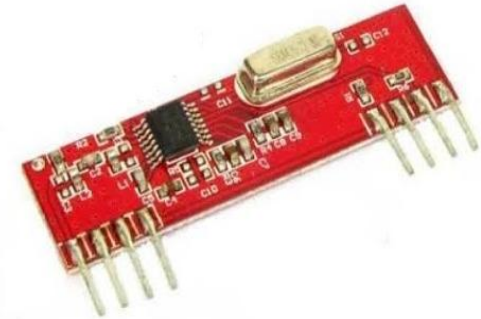
**Components to repel the animals at the
boundary of the farm**

Driver IC

RF Decoder



RF Receiver

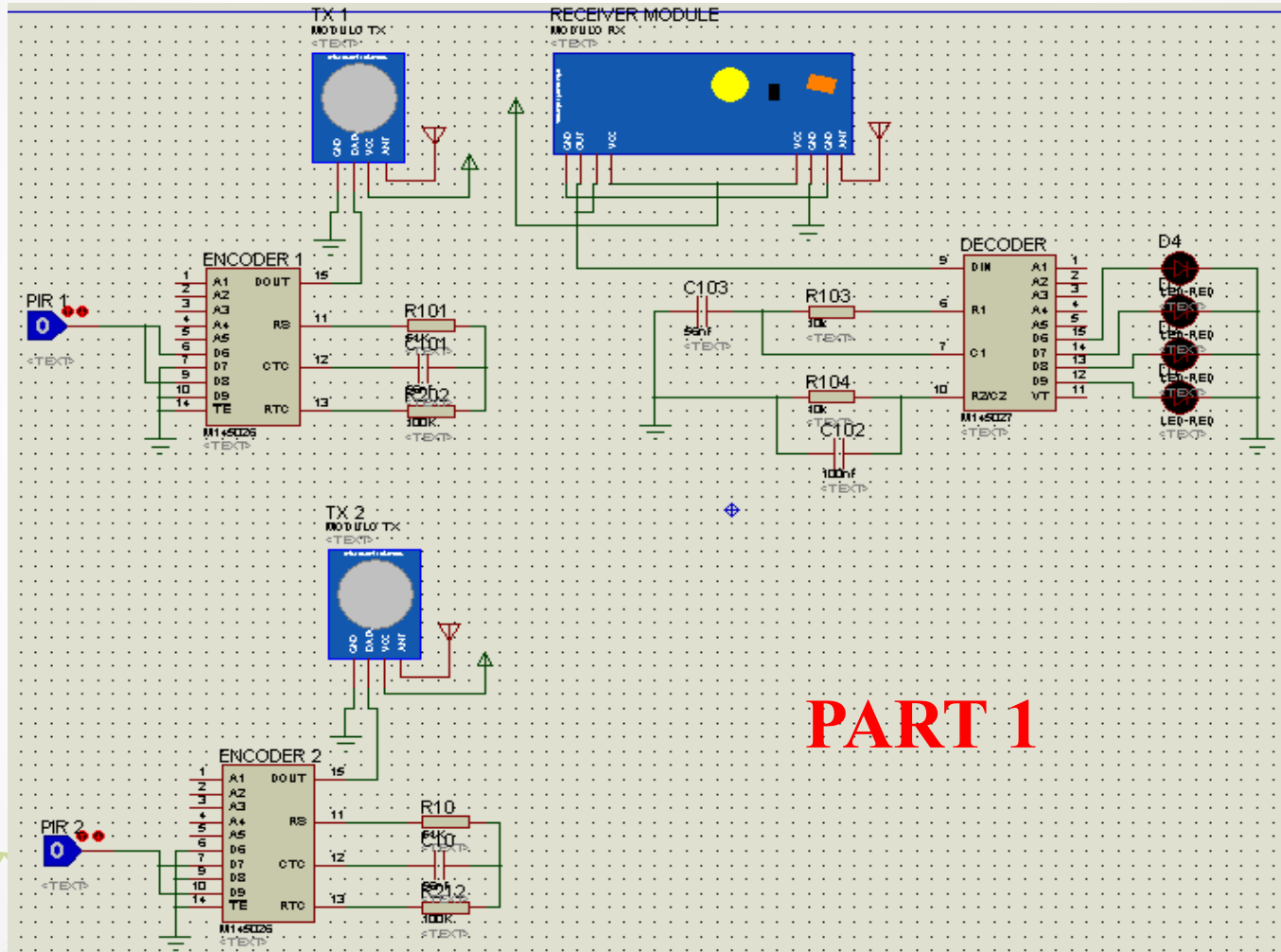


Stepper motor

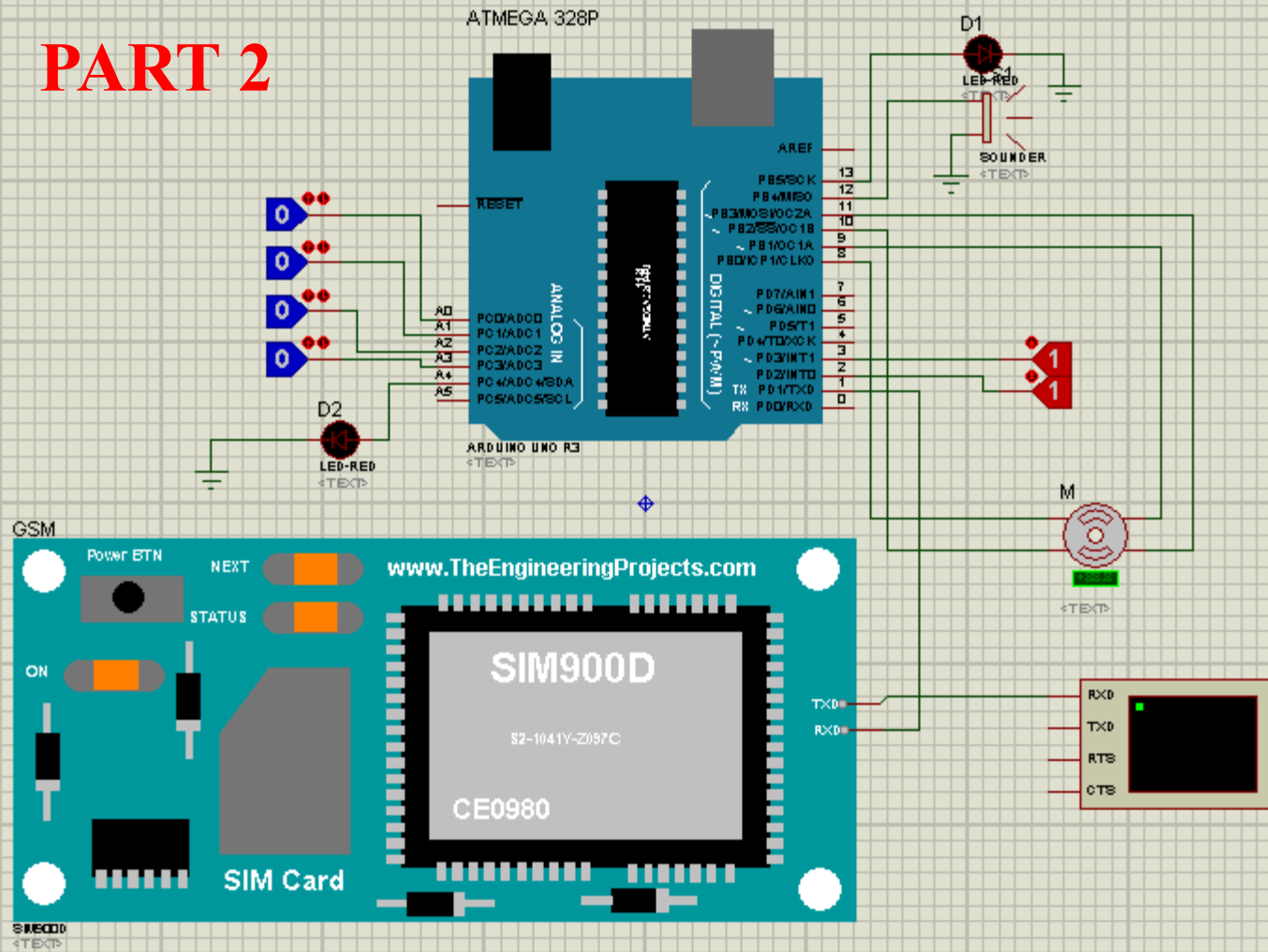


Other components

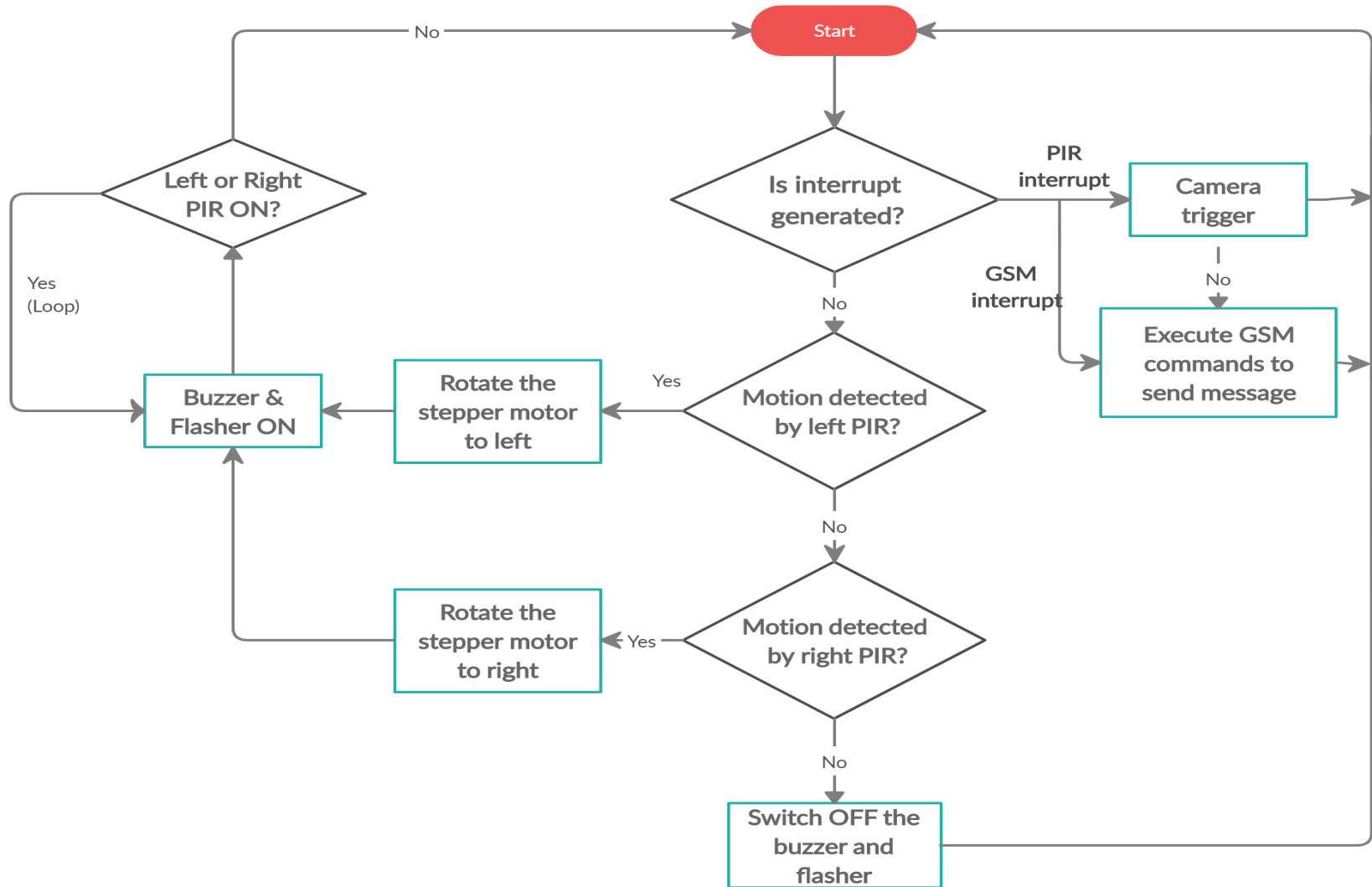
Simulation of the Hardware



PART 2



Programming the Atmega 328p





Digital Image Processing

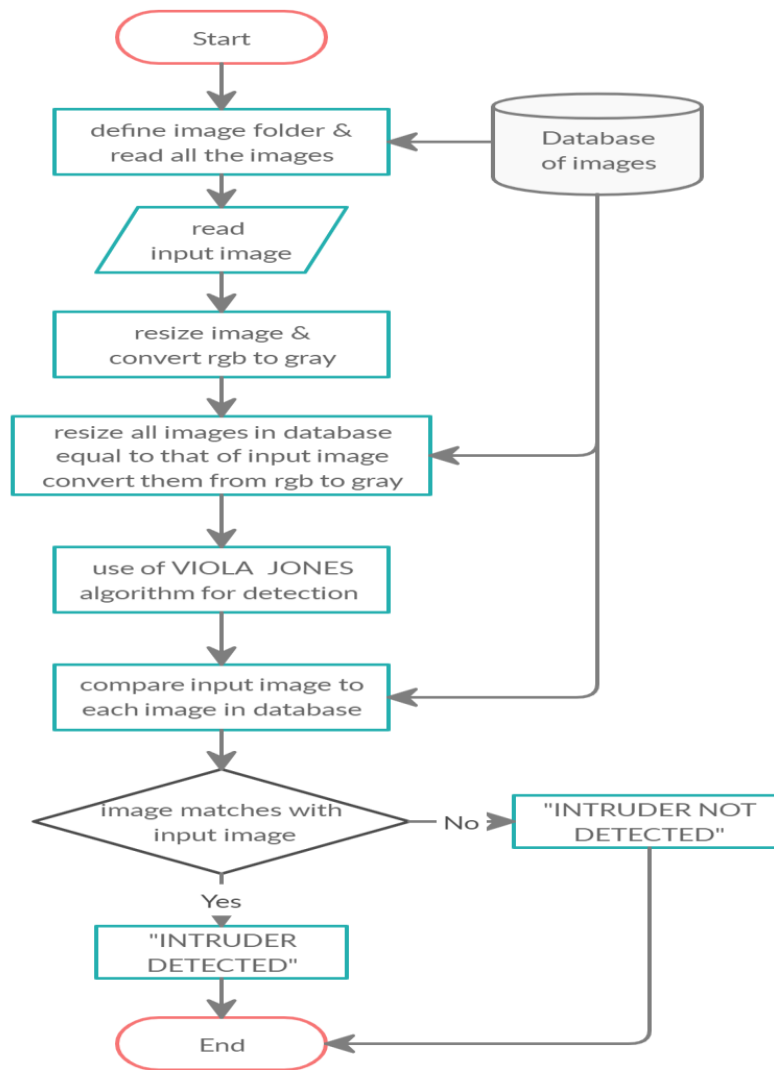


CHALLENGES IN ANIMAL DETECTION

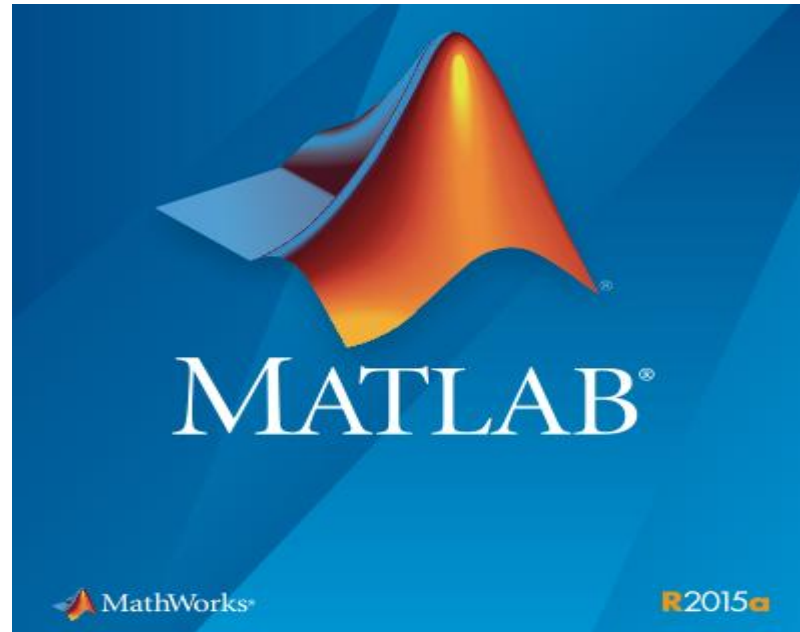
- 1.Variety in animals
- 2.Different body postures
- 3.Light conditions
- 4.Mobility of animals
5. No. of animals(individual/heard)



Flow of the Algorithm

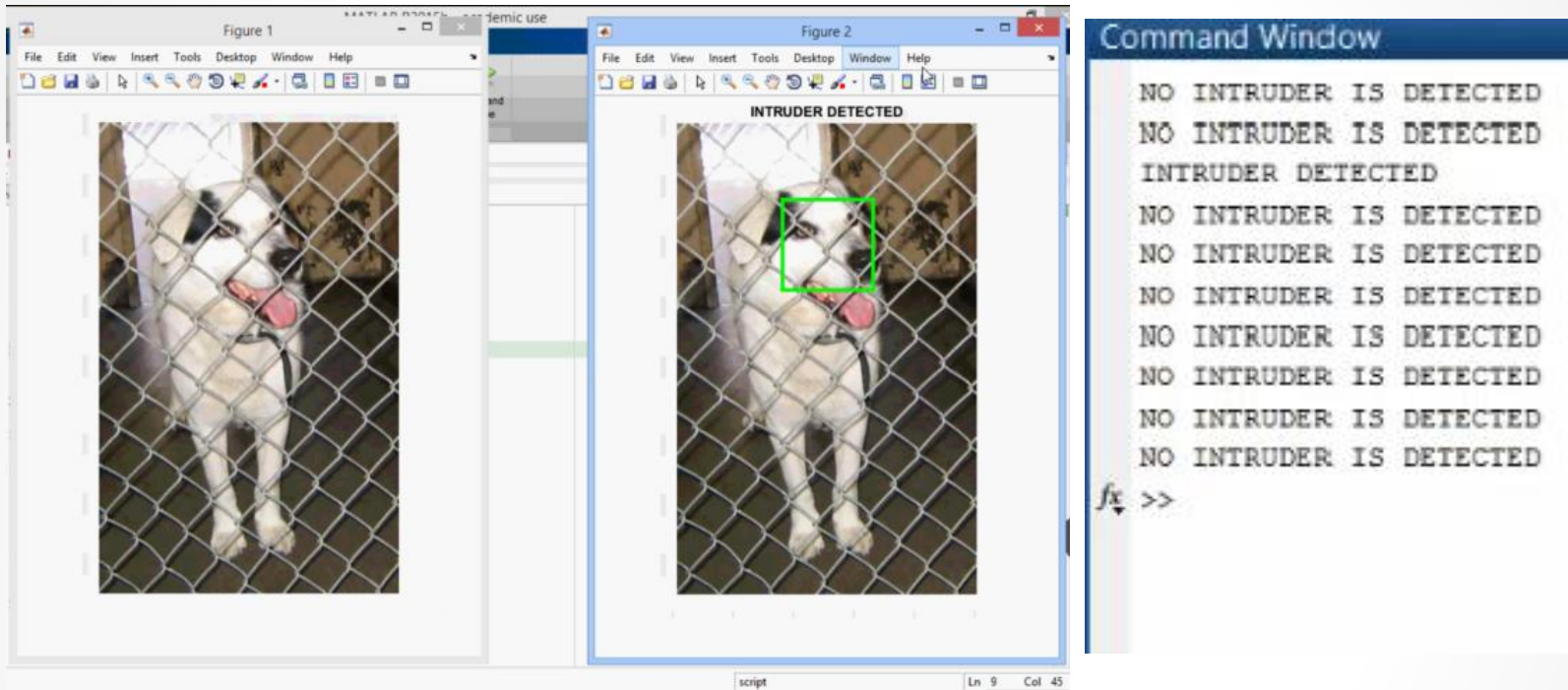


Software used for DIP

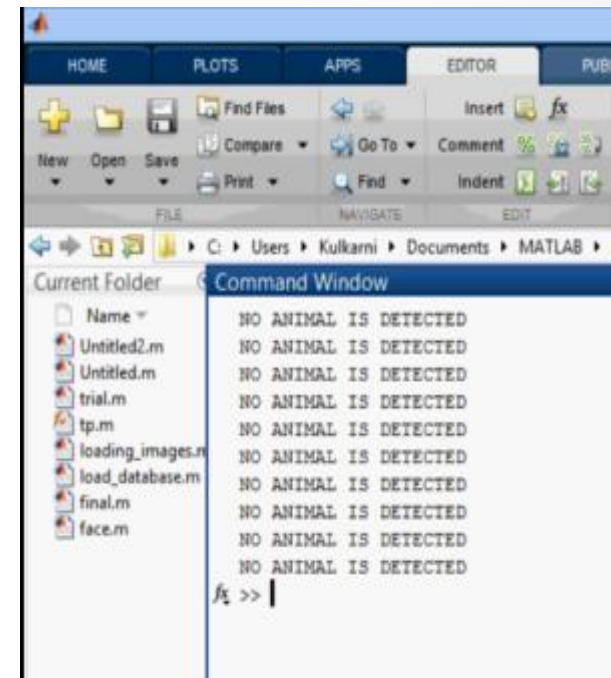
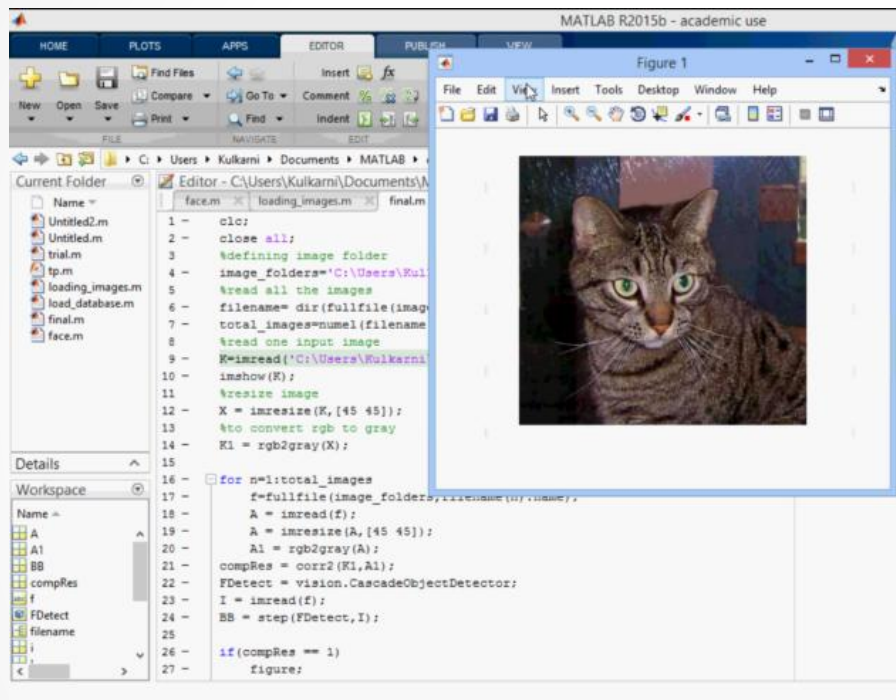


- It is easy to use.
- It is quick to learn due to good documentation.
- It has various toolboxes for functions.

Simulation Results



Positive Input Image



Negative input Image

Work done before lockdown

- Selection of the components
- Formation of the block diagram
- Simulation and implementation of the transmitter and receiver.
- PIR sensor testing
- Collection of database for image processing



Work done in lockdown

- **Analyzing and testing various algorithms**
- **Implementation and testing Viola-Jonnes algorithm**
- **Interfacing of GSM, Stepper motor and other components to microcontroller through simulation**
- **Final Report preparation**
- **Paper presentation**



Work distribution

- **Omkar-** image processing algorithm
paper presentation
- **Anjali-** database collection
paper presentation
coding & simulation
- **Shubham-** LATEX report
coding and simulation
powerpoint presentation



REFERENCES

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- [2].Davide Adami , Fabio Vignoli, “ IoT Solutions for Crop Protection against Wild Animal Attacks” , ResearchGate Conference Paper March 2018
- [3]. Debarati Dutta,“ Laser Security System”, International Journal of Scientific & Engineering Research, Volume 7, Issue 4, April-2016
- [4].Dugyala Karthik, R. Ramesh Babu , “Smart Crop Protection System with Image Capture over IOT” , International Journal of Advanced Information Science and Technology (IJAIST) ISSN: 2319:2682 Vol.6, No.11, November 2017
- [5].Vikhram. B, Revathi. B,Shanmugapriya.R, Sowmiya.S, Pragadeeswaran.G, “Animal Detection System in Farm Areas”, International Journal of Advanced Research in Computer and Communication Engineering, Vol. 6, Issue 3, March 2017



INFERENCES

REFERENCE	INFERENCE
1	GSM INTERFENCING
2	WIRELESS TRANSMISSION OF RECEIVED INPUT
3	LASER
4	IMAGE PROCESSING
5	GSM INTERFENCING
FIELD VISIT	FARM IMPLEMENTATION



Thank
you

